

Complex Computing Problem Report



Name: Hafsa Nouman Siddiqui

Enrolment Number: 02-235222-036

Registration Number: 82117

Program & Semester: BSIT – 7A

Course Instructor: Ma'am Darakhshan Syed

Course: Cyber Security | ITC-411

Table of Contents

1. Introduction of Technology / Algorithms.....	3
2. Design / Implementation	3
2.1. Framework of Proposed Solution.....	3
2.2. Algorithms	4
2.3. Flowchart.....	5
2.4. Screenshots.....	6
3. Recommendations for Optimizing the Model & Comparative Analysis	10
3.1. Comparative Analysis of All Models Used.....	10
3.2. Recommendations for Model Optimization.....	11
4. Conclusion.....	11
References	11

1. Introduction of Technology / Algorithms

Intrusion Detection Systems (IDS) are crucial for identifying malicious activities in network traffic. Machine learning algorithms improve IDS by learning patterns of normal and attack traffic. However, high-dimensional datasets pose challenges such as increased computational cost and risk of overfitting.

This project uses the NF-ToN-IoT dataset, a comprehensive network traffic dataset for cybersecurity research. To address dimensionality issues, we implement a feature selection method based on the Modified Binary Grey Wolf Optimizer (MBGWO), a metaheuristic algorithm inspired by grey wolf social hierarchy and hunting behavior. MBGWO optimizes the subset of features to balance classification accuracy and model simplicity.

We then train classifiers — Support Vector Machine (SVM) and Random Forest (RF) — on the selected features to evaluate the impact on detection performance.

2. Design / Implementation

2.1. Framework of Proposed Solution

The solution pipeline consists of:

- **Data Loading & Preprocessing:** Sampling, cleaning (removal of constant/high-missing features), encoding categorical variables, and scaling numeric features.
- **Feature Selection via MBGWO:** An iterative population-based search algorithm that selects subsets of features by maximizing a fitness function combining weighted F1 score and feature count penalty.
- **Model Training & Evaluation:** Training SVM (with PCA for dimensionality reduction) and RF classifiers on the selected features; evaluation using accuracy, precision, recall, and F1 metrics.
- **Artifact Management:** Saving models, encoders, scalers, feature masks, and training history for reuse.
- **Inference Interface:** Streamlit-based GUI for batch and single-sample predictions using saved artifacts.

The overall workflow enables efficient IDS model development while optimizing feature dimensionality.

2.2. Algorithms

Modified Binary Grey Wolf Optimizer (MBGWO):

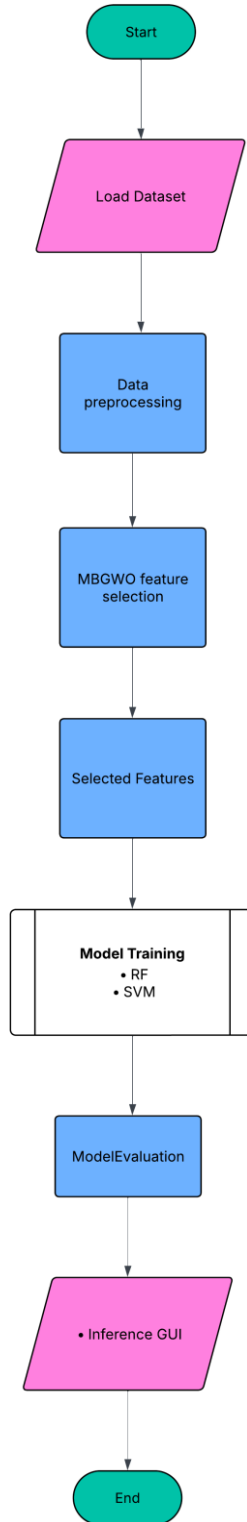
MBGWO simulates grey wolf hunting behavior for binary feature selection:

- **Population Initialization:** Wolves represented by continuous position vectors of feature dimension; passed through sigmoid to generate binary masks.
- **Fitness Function:** Combines weighted F1 score of a Random Forest classifier on a validation split with a penalty proportional to the number of selected features, controlled by parameter α (alpha_w).
- **Hierarchy:** Three best solutions named Alpha, Beta, Delta guide the search.
- **Position Update:** Each wolf updates its position influenced by Alpha, Beta, and Delta wolves via stochastic equations.
- **Early Stopping:** Search terminates if no improvement occurs over a patience number of iterations.

Model Training and Evaluation

- **Support Vector Machine (SVM):** Applied with Radial Basis Function (RBF) kernel. Prior to training, Principal Component Analysis (PCA) reduces dimensionality of the selected features to improve training speed.
- **Random Forest (RF):** An ensemble of decision trees trained on the selected features without dimensionality reduction.
- **Metrics:** Models evaluated on hold-out test data for accuracy, precision, recall, weighted F1-score, and confusion matrices.

2.3. Flowchart



2.4. Screenshots

```
=====
MBGWO Feature Selection
Date: 2025-12-07 18:40:30.939272
=====
Loading dataset from Downloads/NF-ToN-IoT-v3.csv ...
Total rows in dataset: 27520260
Using 1100810 rows (4.0000% of the original dataset)
Detected classes (10): ['Backdoor', 'Benign', 'ddos', 'dos', 'injection', 'mitm', 'password', 'ransomware', 'scanning', 'xss']
Iter 1/50 | best 0.9666 | sel 23/54
Iter 2/50 | best 0.9666 | sel 19/54
Iter 3/50 | best 0.9685 | sel 21/54
Iter 4/50 | best 0.9703 | sel 17/54
Iter 5/50 | best 0.9703 | sel 15/54
Iter 6/50 | best 0.9722 | sel 15/54
Iter 7/50 | best 0.9741 | sel 17/54
Iter 8/50 | best 0.9741 | sel 19/54
Iter 9/50 | best 0.9741 | sel 19/54
Iter 10/50 | best 0.9741 | sel 18/54
Early stopping at iter 10 (no improv for 3 iters)
Iter 01 | Best 0.9666 | Sel 23
Iter 02 | Best 0.9666 | Sel 19
Iter 03 | Best 0.9685 | Sel 21
Iter 04 | Best 0.9703 | Sel 17
Iter 05 | Best 0.9703 | Sel 15
Iter 06 | Best 0.9722 | Sel 15
Iter 07 | Best 0.9741 | Sel 17
Iter 08 | Best 0.9741 | Sel 19
Iter 09 | Best 0.9741 | Sel 19
Iter 10 | Best 0.9741 | Sel 18

Final selected features: 18 / 54 (33.33%)
Selected features (names):
- FLOW_END_MILLISECONDS
- IPV4_DST_ADDR
- FLOW_DURATION_MILLISECONDS
- DURATION_OUT
- MIN_TTL
- LONGEST_FLOW_PKT
- RETRANSMITTED_OUT_BYTES
- RETRANSMITTED_OUT_PKTS
- SRC_TO_DST_AVG_THROUGHPUT
- NUM_PKTS_UP_TO_128_BYTES
- NUM_PKTS_128_TO_256_BYTES
- NUM_PKTS_512_TO_1024_BYTES
- NUM_PKTS_1024_TO_1514_BYTES
- ICMP_TYPE
- DNS_TTL_ANSWER
- DST_TO_SRC_IAT_MIN
- DST_TO_SRC_IAT_AVG
- Label
```

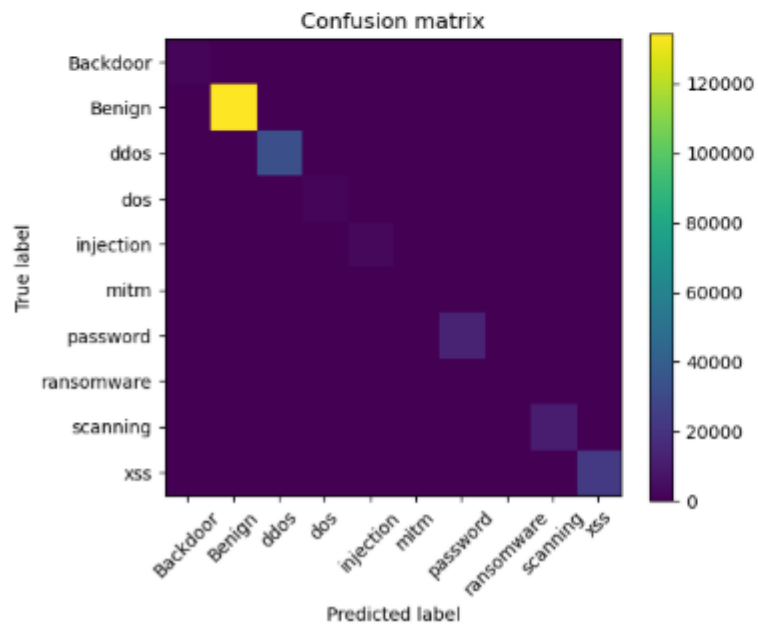
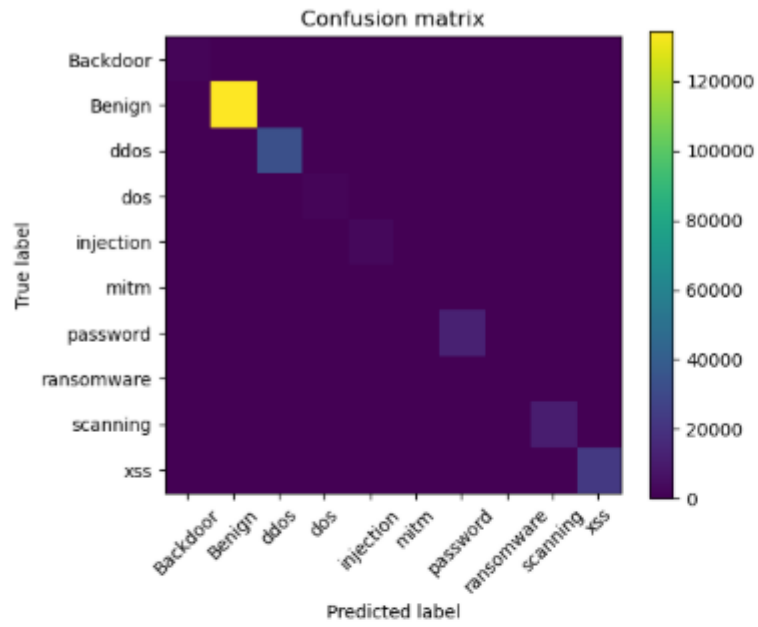
Applying PCA with n_components=18 on data with shape (40000, 18)
Training SVM on 40000 samples and 18 PCA components...

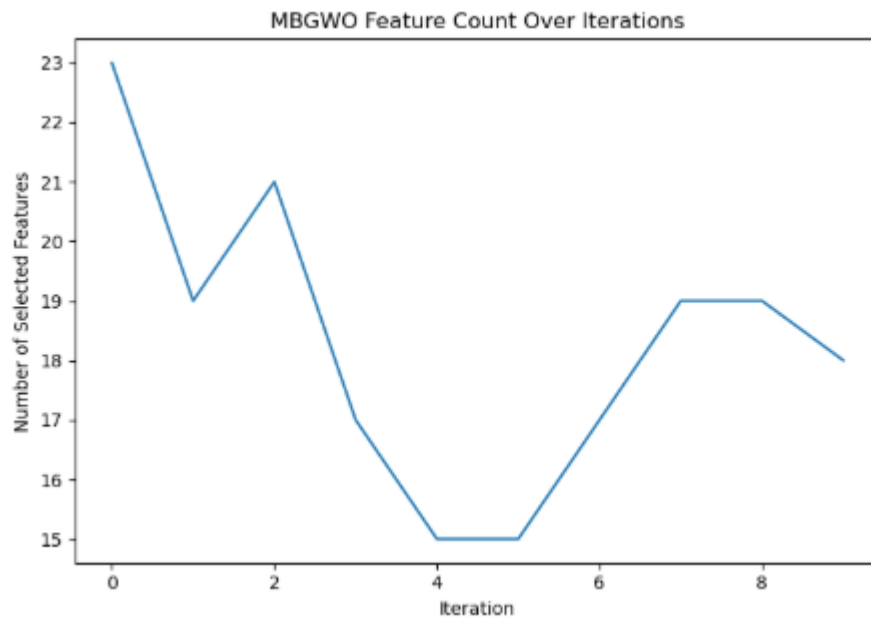
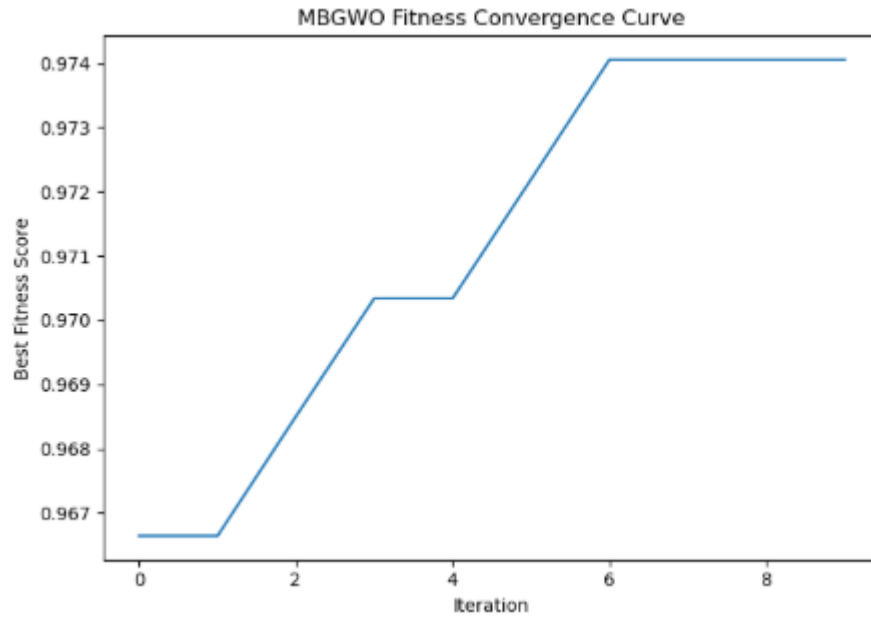
SVM Classification Report:

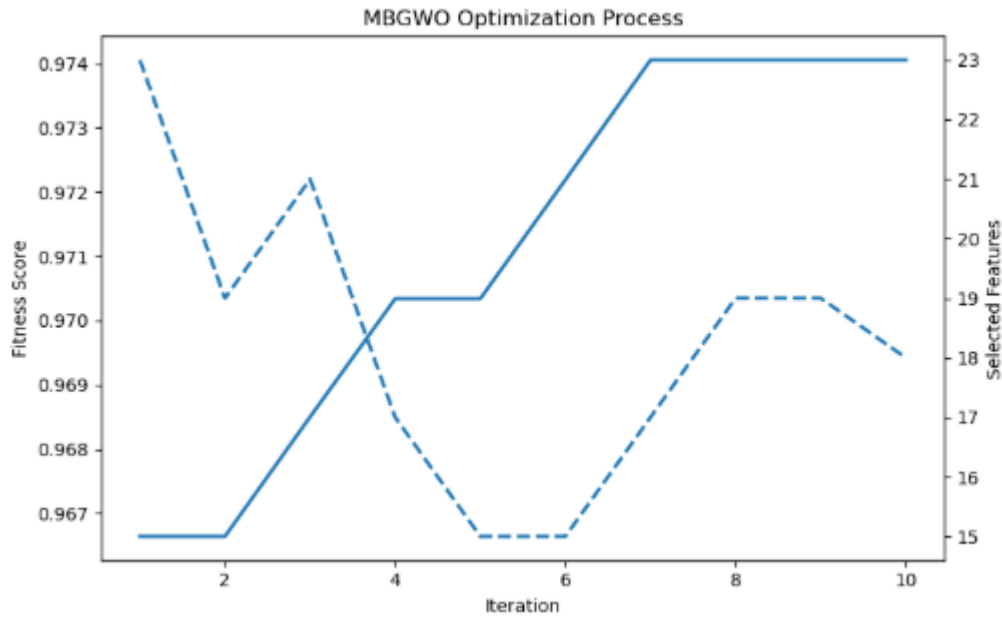
	precision	recall	f1-score	support
0	0.995172	0.999394	0.997279	1650.000000
1	0.998982	0.999866	0.999424	134467.000000
2	0.999051	0.989274	0.994139	33003.000000
3	0.954491	0.995628	0.974625	1601.000000
4	0.909258	0.969281	0.938311	3060.000000
5	0.970588	0.733333	0.835443	45.000000
6	0.983764	0.975232	0.979480	12799.000000
7	0.600000	0.454545	0.517241	33.000000
8	0.999723	0.999078	0.999401	10849.000000
9	0.986486	0.989230	0.987856	22655.000000
accuracy	0.995117	0.995117	0.995117	0.995117
macro avg	0.939752	0.910486	0.922320	220162.000000
weighted avg	0.995194	0.995117	0.995129	220162.000000

Random Forest Classification Report:

	precision	recall	f1-score	support
0	0.999394	0.998788	0.999091	1650.000000
1	1.000000	1.000000	1.000000	134467.000000
2	1.000000	1.000000	1.000000	33003.000000
3	0.999375	0.998751	0.999063	1601.000000
4	0.999673	1.000000	0.999837	3060.000000
5	1.000000	1.000000	1.000000	45.000000
6	1.000000	0.999922	0.999961	12799.000000
7	0.969697	0.969697	0.969697	33.000000
8	0.999908	0.999908	0.999908	10849.000000
9	0.999912	1.000000	0.999956	22655.000000
accuracy	0.999968	0.999968	0.999968	0.999968
macro avg	0.996796	0.996707	0.996751	220162.000000
weighted avg	0.999968	0.999968	0.999968	220162.000000







3. Recommendations for Optimizing the Model & Comparative Analysis

3.1. Comparative Analysis of All Models Used

Model	Features Used	Accuracy	Precision	Recall	F1-Score	Notes
SVM + PCA	MBGWO-selected subset (dynamic PCA components)	0.9951	0.9952	0.9951	0.9951	PCA helps reduce dimensionality for SVM training.
Random Forest	MBGWO-selected subset	0.99997	0.99997	0.99997	0.99997	RF can handle selected features directly.

- Feature selection via MBGWO significantly reduces feature count while maintaining or improving classification performance.
- SVM benefits from PCA due to sensitivity to high dimensionality.
- RF tends to be more robust but slower when using many features.

3.2. Recommendations for Model Optimization

- **MBGWO Tuning:** Experiment with population size, iterations, and α parameter to balance feature sparsity and accuracy.
- **Subsampling:** Using `subsample_ratio < 1` during MBGWO fitness evaluation speeds up optimization.
- **Ensemble Methods:** Combine multiple models or feature selectors for better robustness.
- **Advanced Preprocessing:** Investigate feature engineering and handling class imbalance.
- **Hyperparameter Tuning:** Use GridSearch or RandomizedSearch for SVM and RF parameters.
- **Incremental PCA or other DR:** For large datasets, incremental PCA or manifold learning could be considered.
- **Cross-validation:** Use k-fold validation to better estimate model generalization.

4. Conclusion

This project demonstrates effective feature selection on a complex IDS dataset using the MBGWO metaheuristic, significantly reducing dimensionality while preserving classification accuracy. The trained SVM and Random Forest models on the selected features show promising results for intrusion detection tasks. The pipeline including artifact saving and a user-friendly Streamlit inference GUI facilitates practical deployment.

Future work can focus on further hyperparameter optimization, testing on other IDS datasets, and integrating more advanced models.

References

- [1] The University of Queensland Australia, "Machine Learning-Based NIDS Datasets," [Online]. Available: https://staff.itee.uq.edu.au/marius/NIDS_datasets/.

